W-L-126a
FEBRUARY 27, 1959
SUPERSEDING
Int. Fed. Spec. W-L-00126 (COM-NBS)
August 25, 1954

FEDERAL SPECIFICATION

LAMP, INCANDESCENT, PROJECTION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 Sizes.—Lamps covered by this specification are projection lamps of the sizes listed in table I and figures 1 through 6.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

UU-T-111 — Tape; Paper, Gummed (Sealing and Securing).

PPP-B-566 — Boxes, Folding, Paperboard.

PPP-B-585—Boxes, Wood, Wirebound.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Boxes, Fiber.

PPP-B-676 — Boxes, Set-Up, Paperboard.

PPP-P-291 — Paperboard, Wrapping, Cushioning.

Federal Standards:

Fed. Std. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. 123 — Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following publications, of the issues in effect on date of invitation for bids, form a part of this specification:

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American Standards Association, Inc. Standards:

C78.370–1956 — American Standard Code for the Designation of Photo Lamps.

(Application for copies should be addressed to the American Standards Association, Inc., 70 East Forty-fifth Street, New York 14, N. Y.)

Official Classification Committee Regulations:

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

American Trucking Associations, Inc.:

National Motor Freight Classification Rules and Container Specifications for Truck Shipments.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1424 Sixteenth Street, NW., Washington 6, D. C.)

3. REQUIREMENTS

3.1 General.—All lamps shall conform to the standard shapes and sizes of bulbs, forms of filament, types of bases, and other technical characteristics indicated in table I and figures 1 through 6.

3.1.1 *Design voltage.*—The design voltage of all projection lamps referenced in this specification shall be 118 volts. (See 6.1.)

3.1.2 *Range voltage.*—The range voltage of all projection lamps referenced in this specification shall be designated 115-120 volts.

3.2 Minimum initial luminous flux.—The lamps shall comply with the requirements of 4.4.2.

3.2.1 *Maintenance of total luminous flux.*—The lamps shall comply with the requirements of 4.5.6.

3.2.2 *Life.*—The lamps shall comply with the requirements of 4.5.4.

3.3 Identification marking.—Markings shall be in accordance with figures 1 through 6, as applicable, and shall be placed either on the base of each lamp or on the bulb itself, whichever method lends itself to more permanent legibility.

TABLE I.

WATTS	ASA Code	Bulb	Base	Filament	Cooling	Figure
100	CDX or CDS	T–8	S–C–Bay	CC–13	Convection	1
150	CFC or CEW	T–8	S–C–Bay	2CC–8	Convection	1
200	CVX	T–10	Med Pf.	CC–13	Convection	4
300	CMT	T–8½	S–C–Bay	C–13D (Bi Plane)	Forced	1
300	CXK	T–10	Med Pf.	C–13	Convection	4
500	CZX	T–10	Med Pf.	C–13D	Forced	4
500	DMX	T–20	Med Pf.	C–13	Convection	3
750	DEC	T–12	D–C Med–Pf Ring*	C–13D	Forced	5
750	DDB	T–12	Med Pf.	C–13D	Forced	4
750	DEJ	T–12	Special Pf.	C–13D	Forced	6
1000	DSB	T–20	Mog. Pf.	C–13D	Convection	2
1000	DFD	T–12	Med Pf.	C–13D	Forced	4
1000	DRS	T–20	Med Pf.	C–13D	Forced	3
1000	DGS	T–12	D–C Med–Pf Ring*	C–13D	Forced	5
1000	DFK	T–12	Special Pf.	C–13D	Forced	6

* NOTE: Lamps DEC and DGS Burn Base-up all others Burn Base-down.

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3.4 Workmanship.—Workmanship shall be of the highest grade throughout. All metal parts shall be free from major chips, burns, and scratches in accordance with good commercial practice.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Test procedure. — Unless otherwise specified, all tests shall be made in a competent and expert engineering manner by the manufacturer. The manufacturer will be required to supply the necessary equipment, assistance, power, and facilities for making such initial tests and examinations and required to keep proper records of examinations and tests. The procuring activity shall have the privilege of witnessing and verifying all tests of the lamps made hereunder, and shall also have the privilege of having access to the records of such tests at all reasonable times, the method of test procedure may be modified in any particular provided that it is mutually agreed that such modified procedure will yield results equivalent to those obtained with the procedure specified herein.

4.2 Submission for inspection. — Lamps shall not be submitted for inspection prior to 48 hours from the time that they are manufactured.

4.3 Lot definition.—A lot shall be all of the lamps of any type and size offered for acceptance at any one time.

4.4 Acceptance inspection. — Projection lamps shall be subjected to physical inspection tests and initial rating tests in accordance with the sampling procedures of Military Standard MIL–STD–105. Acceptance quality levels (A.Q.L.'s) shall be 1.0 percent for major defects and 4.0 percent for minor defects.

4.4.1 *Rating test conditions.*—Before subjecting lamps to the initial rating test they shall be sufficiently burned or seasoned to have reached stable values of watts and lumens. Lamps shall be operated at 118 volts plus or minus 0.5 volt. The measurements of total lumens shall be made with an accurate photometric device capable of accepting and integrating the entire luminous output of the lamp under test, such as a four foot diameter integrating sphere.

4.4.2 *Rejection for low initial lumens.*—Any lot of lamps shall be rejected if 20 percent + two or more of the rating test lamps fall below the value listed in column headed "Minimum total lumens initial" in table II.

4.5 Life performance test lamp selection and test quantities.—The lamps of the life performance test shall be selected from the rating test lamps at the ratio of not less than 5 from the first 1500 lamps of any lot and at least 1 lamp from each 500 lamps (or part thereof) by which the lot exceeds 1500 lamps. The life-test lamps selected shall be representative of the rating-test lamps.

4.5.1 *Life-tested voltage.*—Life-test lamps shall be operated at 118 volts plus or minus 0.5 volt.

4.5.2 *Position of burning.*—Lamps shall be operated in the position of burning shown in table I and figures 1 through 6.

4.5.3 *Lamp life.*—The life of a lamp is the number of hours life of the lamp to burn out.

4.5.4 *Rejection for failure to meet life requirement.*—Any lot of lamps, provided that such lot is represented on test by the minimum required number of lamps (not less than 5) selected in accordance with 4.5 shall be rejected if the average life of the life-test lamps falls below the life value given in the applicable figure by more than the life tolerance specified in table III for the number of life-test lamps averaged.

4.5.5 *Determination of maintenance of total lumens.* — When life-test lamps have

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TABLE II.—*Light output*

WATTS	ASA Code	Minimum total lumens		
		Initial		At 70 percent of rated life
		Clear	Opaque	Clear
100	CDX or CDS	1,650		
150	CFC or CEW	2,800		2,450
300	CMT	6,200		5,450
200	CVX	3,650	3,360	3,200
300	CXK	6,550	6,320	5,650
500	CZX	11,400	10,550	9,700
750	DDB	18,400	17,000	15,500
1000	DFD	26,600	24,900	21,500
500	DMX	11,400		9,100
1000	DRS	24,100		19,500
1000	DSB	24,100		19,500
750	DEJ	18,400	17,000	15,500
1000	DFK	26,600	24,900	21,500
750	DEC	18,400		15,500
1000	DGS	26,600		21,500

TABLE III.—*Life tolerance*[1]

Number of lamps, averaged	Allowable percent variation from rated life	Number of lamps averaged	Allowable percent variation from rated life	Number of lamps averaged	Allowable percent variation from rated life
250 and above	5	24–20	17	9	25
249–100	8	19–18	18	8	27
99–55	10	17–16	19	7	28
54–45	11	15–14	20	6	31
44–35	13	13–12	22	5	34
34–30	14	11	23		
29–25	15	10	24		

[1] On account of the natural and inherent variations in individual lamp performance, the average test results of a small number of test samples cannot be depended upon to indicate exactly the average from the larger quantity from which the test samples were taken. The tables of allowable variations are devised in accordance with the observed accuracy of lamp testing methods and the observed variations in lamp performance.

burned for 70 percent of rated average life, they shall be measured for maintenance of total lumens under the conditions of 4.4.1.

4.5.6 *Rejection for failure to meet maintenance requirements.* — Any lot of lamps shall be rejected provided such lot is represented on test by the minimum required number of lamps (not less than 5) in accordance with 4.5 if the average total lumens after 70 percent of rated average life, determined in accordance with 4.5.5 falls below the value specified in table II by more than the tolerance specified in table IV.

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TABLE IV.—*Maintenance tolerances*[1]

Number of lamps tested	Allowable percent variation
100 and above	2
99–25	4
24–10	6
9–5	8

[1] On account of the natural and inherent variations in individual lamp performance, the average test results of a small number of test samples cannot be depended upon to indicate exactly the average for the larger quantity from which the test samples were taken. The tables of allowable variation are devised in accordance with the observed accuracy of lamp-testing methods and the observed variations in lamp performance.

4.5.7 *Life-test conditions.*—Lamps shall be burned intermittently; 15 minutes on, 5 minutes off. Lamps shall be burned in lamp houses equipped with spherical mirrors and lenses positioned at the level of the light center and spaced at least ¼ inch from the nearest bulb surface. Lamp houses shall be provided with either convection or forced ventilation as required by the individual lamp type as shown in table I.

5. PREPARATION FOR DELIVERY

For civil agency procurement the levels of preservation, packaging, and packing protection used are defined in Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A.*—Projection lamps shall be packaged in accordance with the procedures specified for the designated methods as described in the following paragraphs.

5.1.1.1 *Projection lamps.*—Package each lamp individually, method III as follows: Cushion each lamp within flexible, single-face corrugated fiberboard, type I, conforming to Federal Specification PPP–P–291. Secure the cushioning with gummed paper tape, class I, conforming to Federal Specification UU–T–111. Place each cushioned lamp within a folding box conforming to Federal Specification PPP–B–566 or PPP–B–676. Closure shall be in accordance with the appendix of the box specification. As an alternative, a pre-formed and glued flexible single-face corrugated-fiberboard wrapper may be used, or a carton with an integral internal cushioning may be used.

5.1.1.2 *Intermediate package.* — Place a quantity of packaged lamps within a close-fitting, style RSC, type I, grade 5 fiberboard box conforming to Federal Specification PPP–B–636. In shipments of less than case quantities fill all voids with flexible single-face corrugated fiberboard or excelsior or other cushionable material, to prevent movement. Seal the box in accordance with the appendix of the box specification.

5.1.2 *Level C.*—Projection lamps shall be packaged in accordance with commercial practice.

5.1.3 *Marking.* — Intermediate packages shall bear the following markings in addition to the standard commercial markings: Contract or order number and the Federal identification number which is cited on the applicable figure. Individual lamp cartons will bear at least the following markings: Wattage, voltage, ASA code, and manufacturer's name or trade-mark.

5.2 Packing.

5.2.1 *Level A.*—A quantity of intermediate packages containing projection lamps shall be packed within a wood box conforming to Federal Specification PPP–B–585, PPP–B–601, or PPP–B–621. Fabricate the box to fit the contents snugly. The gross weight of any shipping container shall not exceed 150 pounds. Close the shipping container in accordance with the appendix of the box specification.

5.2.1.1 *Metal strapping.*—Shipping containers shall be strapped in conformance with

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the requirements of the appendix of the container specification.

5.2.1.2 *Level B.*—A quantity of intermediate packages containing projection lamps shall be packed within a wooden box conforming to Federal Specification PPP–B–585, PPP–B–601, or PPP–B–621. Fabricate the box to fit the contents snugly. The gross weight of any shipping container shall not exceed 200 pounds. Closure shall be in accordance with the box specification. Strapping is not required.

5.2.2 *Level C.*—Projection lamps shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification for rail shipment, National Motor Freight Classification for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed.

5.2.3 *Marking (military).*—Intermediate packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Military Standard MIL–STD–129.

5.2.4 *Marking (nonmilitary).* — Unless otherwise specified, exterior shipping containers shall be marked in accordance with Federal Standard No. 123.

6. NOTES

6.1 Ordering data. — Purchasers should specify ASA Code and voltage (115–120 volts). In isolated cases, where a limited and special requirement exists for projection lamps designed for 100, 105, 110, 125 and 130 volt operation (for example, for test or laboratory use), such lamps shall be procured commercially from local funds.

6.1.1 Procurement documents should stipulate that the life-test lamps selected by the inspector and destroyed in the course of the life-test are to be counted as part of the quantity of lamps delivered by the contractor to the Government.

6.2 Transportation description.—Transportation description applicable to these items is:

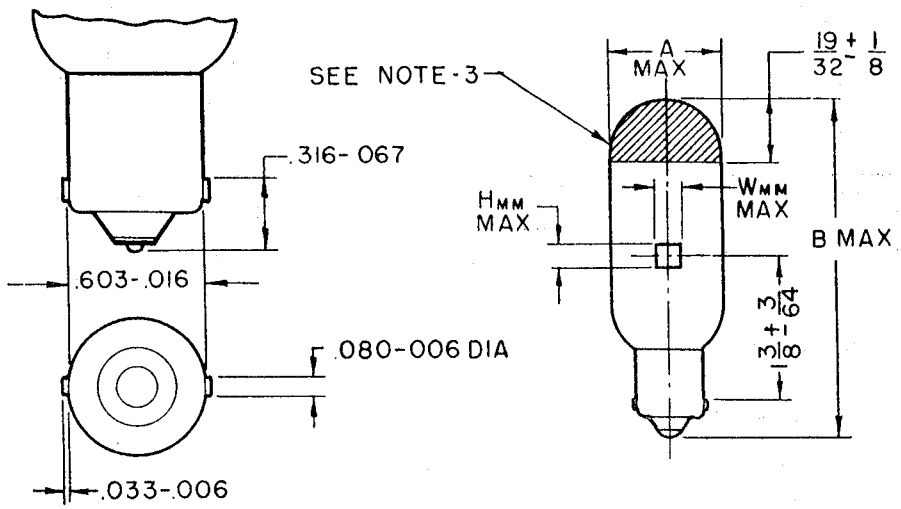
Lamps, Electric, Incandescent, including photoflash.

Carload minimum weight 15,000 and 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor volume minimum weight — any quantity.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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ENLARGED VIEW OF BASE

DIMENSIONS				BULB	ELECTRICAL CHARACTERISTICS			FIIN	ASA
A	B	H	W		WATTS	AVERAGE LIFE IN HOURS	FILAMENT		
1.041	3 1/8	5.5	6.0	T-8	100	50	CC-13	542-6657	CDX OR CDS
1.041	3 5/8	7.0	5.5	T-8	150	25	2CC-8	143-3106	CFC OR CEW
1.104	4 1/8	7.6	7.3	T-8½	300	25	C-13D (BIPLANE)	542-6658	CMT

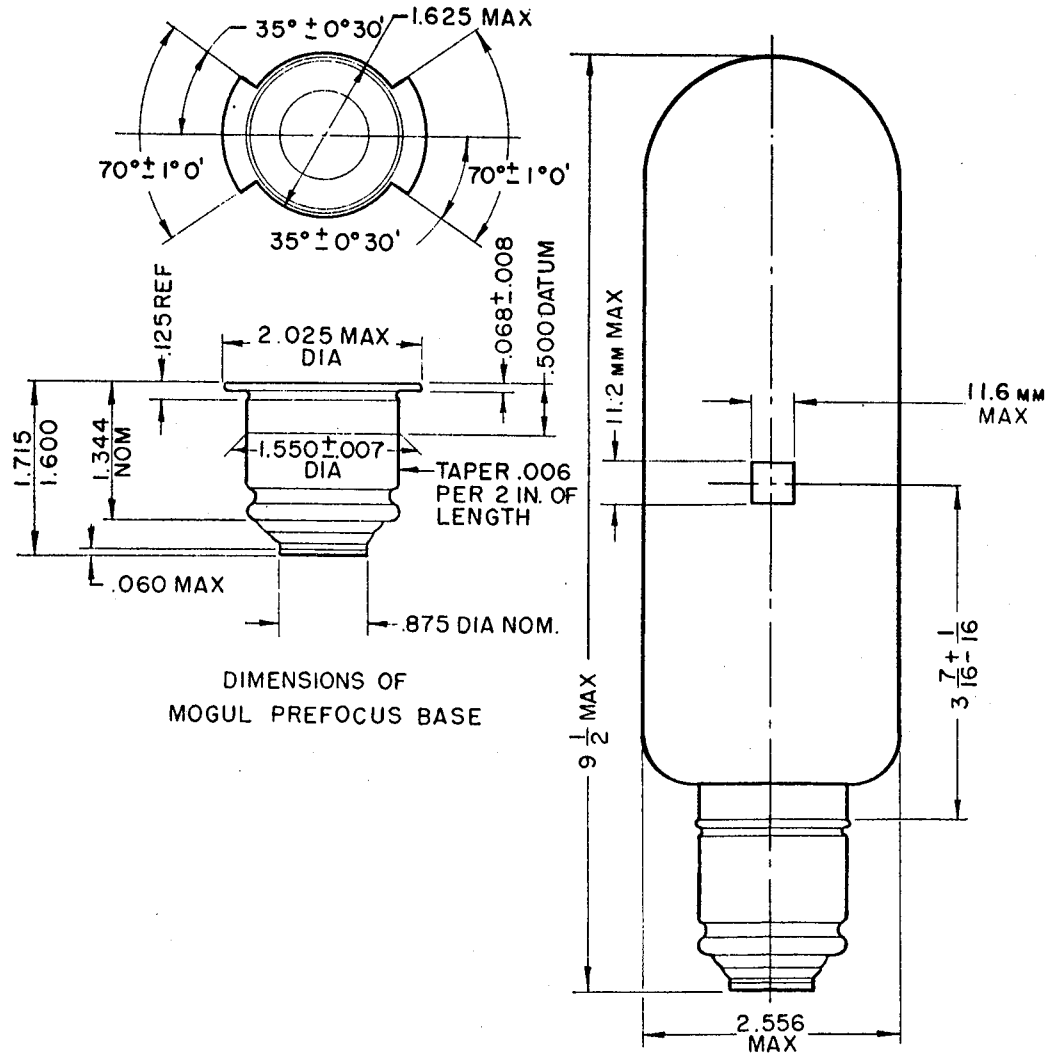
NOTES:

1. DESIGN VOLTAGE OF LAMPS IS 118 VOLTS. RANGE VOLTAGE IS 115-120 VOLTS.
2. LAMPS ARE MARKED IN A CONSPICUOUS PLACE WITH THE FOLLOWING INFORMATION: RANGE VOLTAGE, WATTAGE, MANUFACTURER'S NAME OR CODE DESIGNATING LETTERS AND ASA CODE DESIGNATION.
3. OPAQUE TOPS ARE ACCEPTABLE ON 100 AND 150 WATT LAMPS.

ALL DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED.

FIGURE 1.—Single contact, bayonet, base-down.

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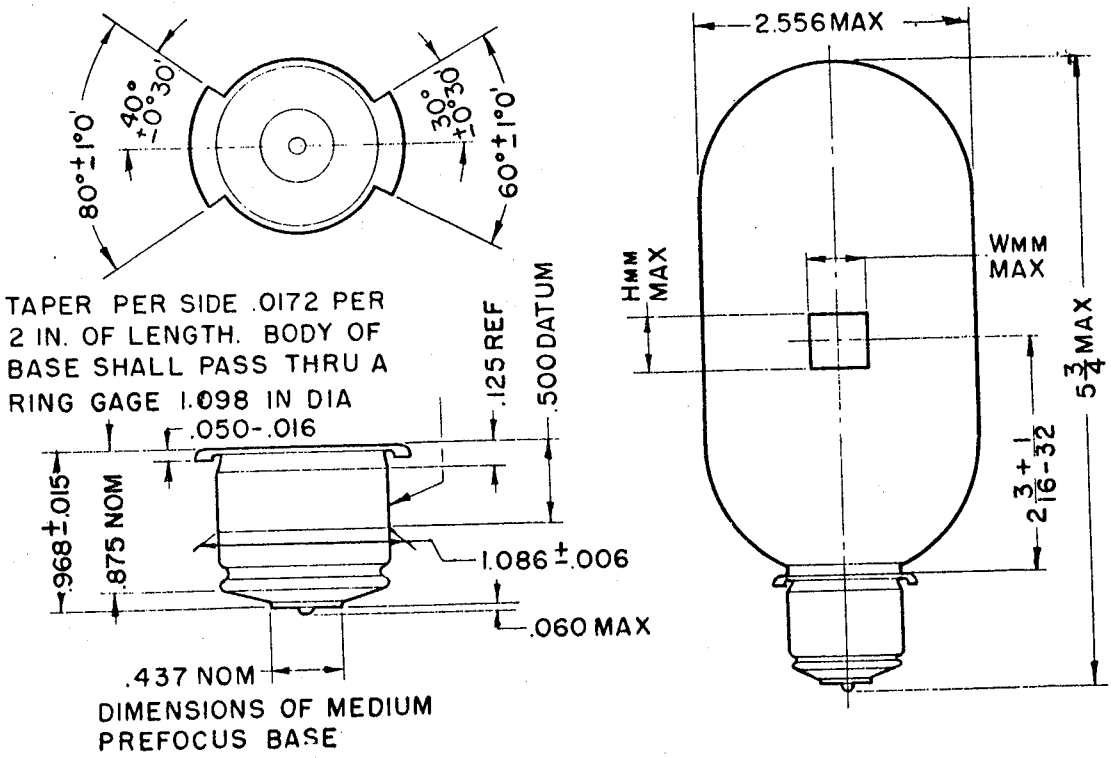
BULB	ELECTRICAL CHARACTERISTICS			FIIN	ASA
	WATTS	AVERAGE LIFE IN HOURS	FILAMENT		
T-20	1000	25	C-13D	408-6058	DSB

NOTES:
1. DESIGN VOLTAGE OF LAMP IS 118 VOLTS. RANGE VOLTAGE IS 115-120 VOLTS.
2. THE LAMP IS MARKED IN A CONSPICUOUS PLACE WITH THE FOLLOWING INFORMATION: RANGE VOLTAGE, WATTAGE, MANUFACTURER'S NAME OR CODE DESIGNATING LETTERS AND ASA CODE DESIGNATION.

ALL DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED.

FIGURE 2.—Mogul prefocus, base-down.

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DIMENSIONS		BULB	ELECTRICAL CHARACTERISTICS			FIIN	ASA
H	W		WATTS	AVERAGE LIFE IN HOURS	FILAMENT		
11.7	13.0	T-20	500	50	C-13	143-7468	DMX
11.2	11.6	T-20	1000	25	C-13D	143-3044	DRS

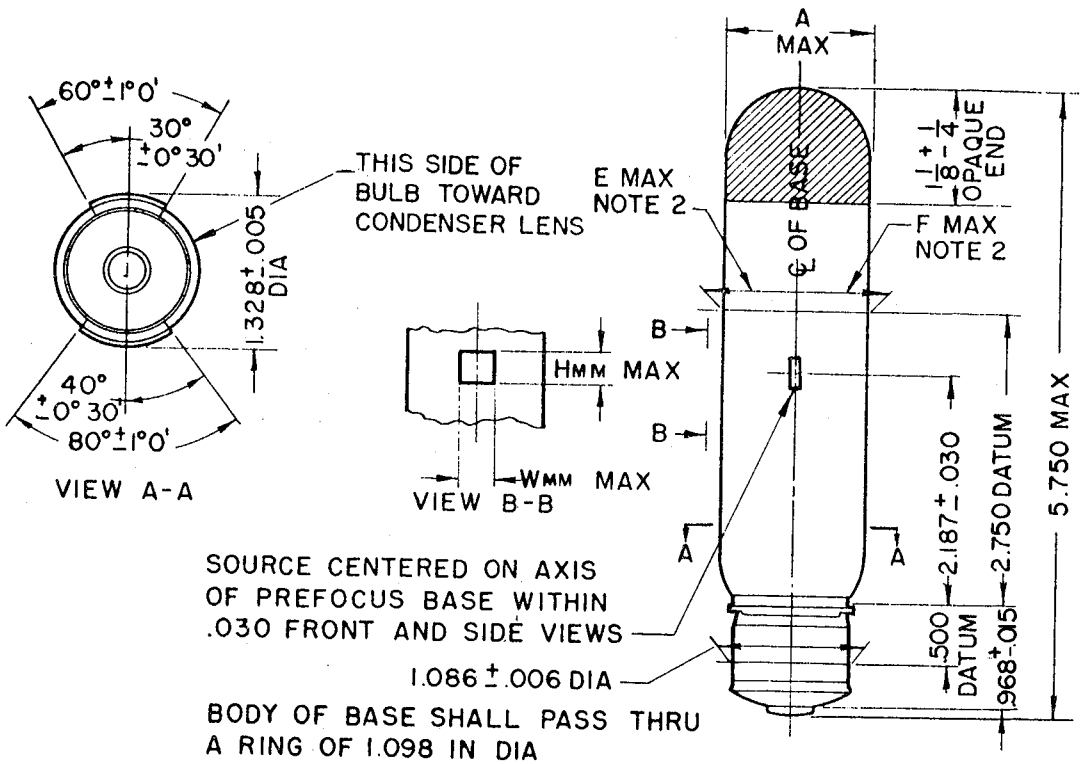
NOTES:

1. DESIGN VOLTAGE OF LAMPS IS 118 VOLTS. RANGE VOLTAGE IS 115-120 VOLTS.
2. LAMPS ARE MARKED IN A CONSPICUOUS PLACE WITH THE FOLLOWING INFORMATION: RANGE VOLTAGE, WATTAGE, MANUFACTURER'S NAME OR CODE DESIGNATION.

ALL DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED.

FIGURE 3.—Medium prefocus, base-down, T–20 bulb.

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DIMENSIONS					BULB	ELECTRICAL CHARACTERISTICS			FIIN	ASA
A	E	F	H	W		WATT	AVERAGE LIFE IN HOURS	FILAMENT		
1.294	.725	.668	7.8	8.5	T-10	200	50	CC-13	542-6658	CVX
1.294	.725	.668	9.3	9.3	T-10	300	25	C-13	143-3040	CXK
1.294	.725	.668	8.6	9.2	T-10	500	25	C-13D	156-1561	CZX
1.546	.850	.785	9.9	9.9	T-12	750	25	C-13D	156-1568	DDB
1.546	.850	.785	10.7	10.7	T-12	1000	10	C-13D	155-8018	DFD

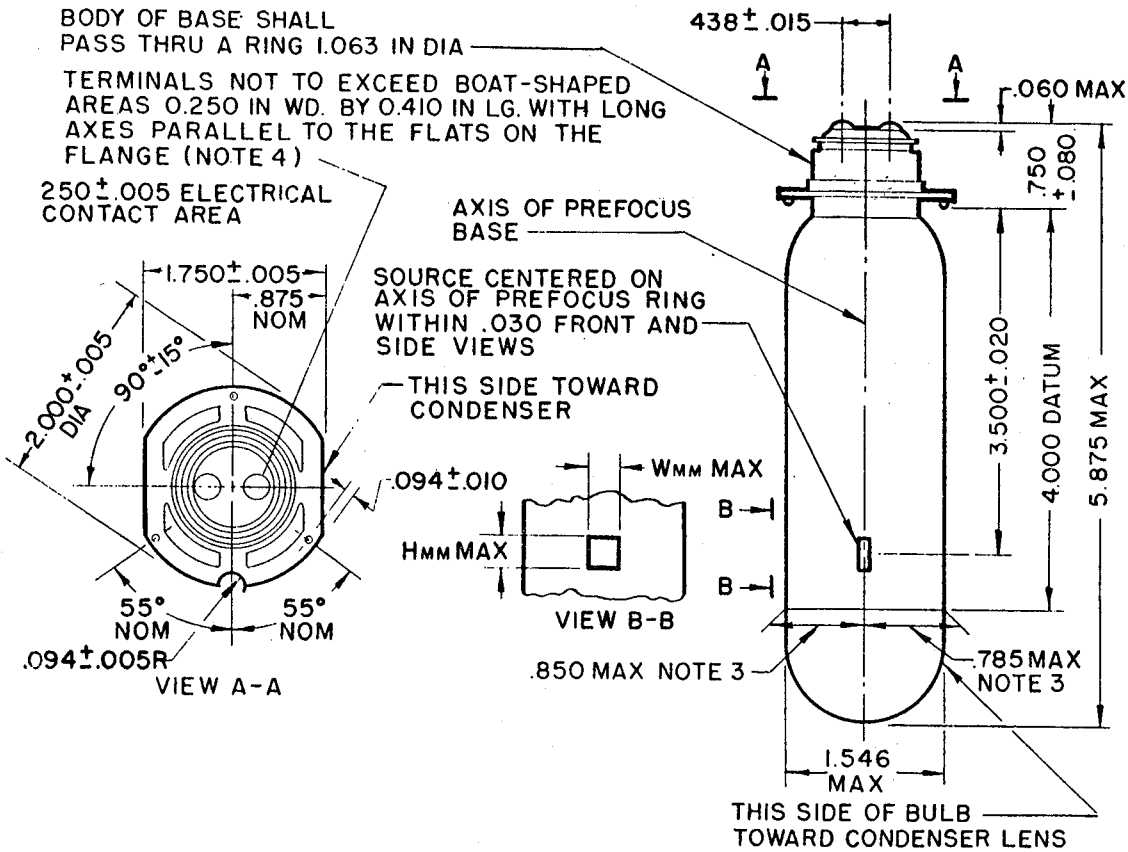
NOTES:

1. DESIGN VOLTAGE OF LAMPS IS 118 VOLTS. RANGE VOLTAGE IS 115 - 120 VOLTS.
2. "E" DIMENSION AND "F" DIMENSION INDICATE THE MAXIMUM EXCURSION OF THE BULB SURFACES FROM THE BASE AXIS TOWARD THE MIRROR AND CONDENSER LENS RESPECTIVELY AT THE POINTS INDICATED, WHEN THE LAMP IS INSERTED IN A HOLDER WHICH ROTATIONALLY POSITIONS THE LAMP AS SHOWN IN VIEW A-A.
3. LAMPS ARE MARKED IN A CONSPICUOUS PLACE WITH THE FOLLOWING INFORMATION: RANGE VOLTAGE, WATTAGE, MANUFACTURER'S NAME OR CODE DESIGNATING LETTERS AND ASA CODE DESIGNATION.

ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED.

FIGURE 4.—Medium prefocus, base-down.

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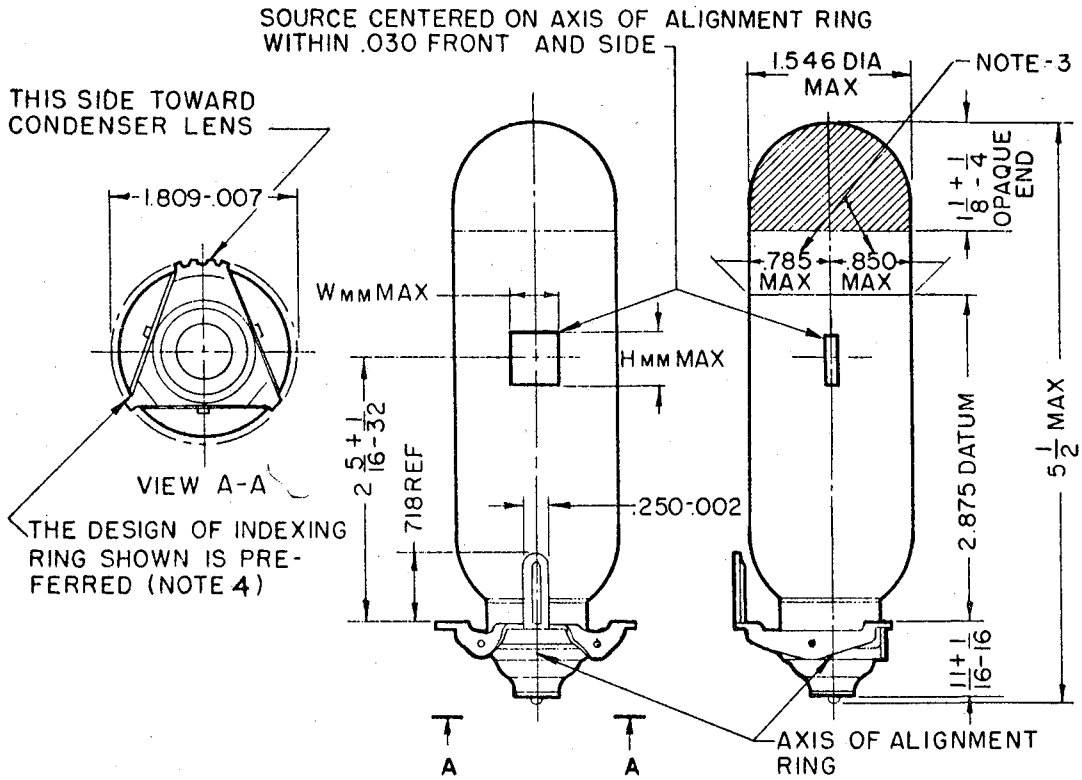
DIMENSIONS		BULB	ELECTRICAL CHARACTERISTICS			FIIN	ASA
H	W		WATTS	AVERAGE LIFE IN HOURS	FILAMENT		
9.9	9.9	T-12	750	25	C-13D	246-5053	DEC
10.7	10.7	T-12	1000	10	C-13D	246-5055	DGS

NOTES:
1. DESIGN VOLTAGE OF LAMPS IS 118 VOLTS. RANGE VOLTAGE IS 115 - 120 VOLTS.
2. LAMPS ARE MARKED IN A CONSPICUOUS PLACE WITH THE FOLLOWING INFORMATION: RANGE VOLTAGE, WATTAGE, MANUFACTURER'S NAME OR CODE DESIGNATING LETTERS, AND ASA CODE DESIGNATION.
3. .850 AND .785 DIMENSIONS INDICATE THE MAXIMUM EXCURSION OF THE BULB SURFACES FROM THE BASE AXIS TOWARD THE MIRROR AND CONDENSER LENS RESPECTIVELY AT THE POINTS INDICATED, WHEN THE LAMP IS INSERTED IN A HOLDER WHICH ROTATIONALLY POSITIONS THE LAMP AS SHOWN IN VIEW A-A.
4. SHAPE OF CONTACTS ARE OPTIONAL, AND DRAWING INDICATES MINIMUM CONTACT AREA.

ALL DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED.

FIGURE 5.—Double contact, medium prefocus ring, base-up.

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DIMENSIONS		BULB	ELECTRICAL CHARACTERISTICS			FIIN	ASA
M	W		WATTS	AVERAGE LIFE IN HOURS	FILAMENT		
9.9	9.9	T-12	750	25	C-13D	233-3470	DEJ
10.7	10.7	T-12	1000	10	C-13D	299-5920	DFK

NOTE:

1. DESIGN VOLTAGE OF LAMPS IS 118 VOLTS. RANGE VOLTAGE IS 115 - 120 VOLTS.
2. LAMPS ARE MARKED IN A CONSPICUOUS PLACE WITH THE FOLLOWING INFORMATION: RANGE VOLTAGE, WATTAGE, MANUFACTURER'S NAME OR CODE DESIGNATING LETTERS, AND ASA CODE DESIGNATION.
3. .850 MAX AND .785 MAX DIMENSIONS INDICATE THE MAXIMUM EXCURSION OF THE BULB SURFACES FROM THE BASE AXIS TOWARD THE MIRROR AND CONDENSER LENS RESPECTIVELY AT THE POINTS INDICATED, WHEN THE LAMP IS INSERTED IN A HOLDER WHICH ROTATIONALLY POSITIONS THE LAMP AS SHOWN IN VIEW A-A.
4. ALTERNATE INTERCHANGEABLE INDEXING RINGS ARE ACCEPTABLE.

ALL DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED.

FIGURE 6.—Special prefocus, base-down.

MILITARY INTERESTS:

Army—Sig E
Navy—Sh
Air Force.

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